



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

2002
HIGHER SCHOOL CERTIFICATE
APRIL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time — 5 minutes
- Working time — 90 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks — 60

- Attempt questions 1–3
- All questions are of equal value

Examiner: D.M.Hespe

Note: This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 60

Attempt Questions 1–3

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

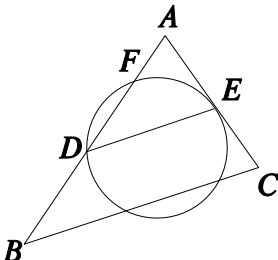
Question 1 (20 marks) Use a SEPARATE writing booklet.

- (a) (i) Calculate $|15 - 20i|$, **1**
- (ii) Find $\arg\left(-1 - \frac{3}{\sqrt{2}}i\right)$. **1**
- (b) Solve $x^2 = 4x - 20$ over the complex field. **2**
- (c) The polynomial $z^3 - 7z^2 + 25z - 39$ has one zero equal to $2 + 3i$. Write down its three linear factors. **3**
- (d) (i) Differentiate $x \ln x$. **1**
- (ii) Hence show that a primitive of $\ln x$ is $x \ln x - x$. **2**
- (e) Give the domain and range of $3 \cos^{-1} 3x$. **2**
- (f) Find the square roots of $5 + 12i$. **2**
- (g) Show that $z - i$ is a factor of $z^3 + 2iz^2 + 3i$. **1**
- (h) Prove that the complex numbers $1 + 6i$, $3 + 10i$, and $4 + 12i$ are collinear on an Argand diagram. **3**
- (i) The curves $y = \ln x$ and $y = x - 1.1$ intersect near $x = 1.5$. Use one iteration of Newton's method to get a better estimate. **2**

Question 2 (20 marks) Use a SEPARATE writing booklet.

(a) Prove that $\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$. 3

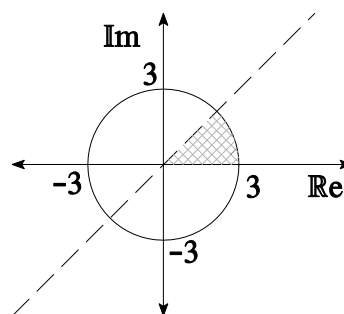
(b) 3



A line DE parallel to the base BC of a triangle ABC cuts AB , AC in D and E respectively. The circle which passes through D , and touches AC at E , meets AB at F . Prove that F , E , C , B lie on a circle.

(c) (i) Sketch the locus of z on an Argand diagram such that $|z| = |z+1-\sqrt{3}i|$. 2

(ii) Describe the locus of z shaded on the diagram. 3



(d) (i) Use De Moivre's theorem to express $\cos 3\theta$ in terms of $\cos \theta$ and $\sin 3\theta$ in terms of $\sin \theta$. 3

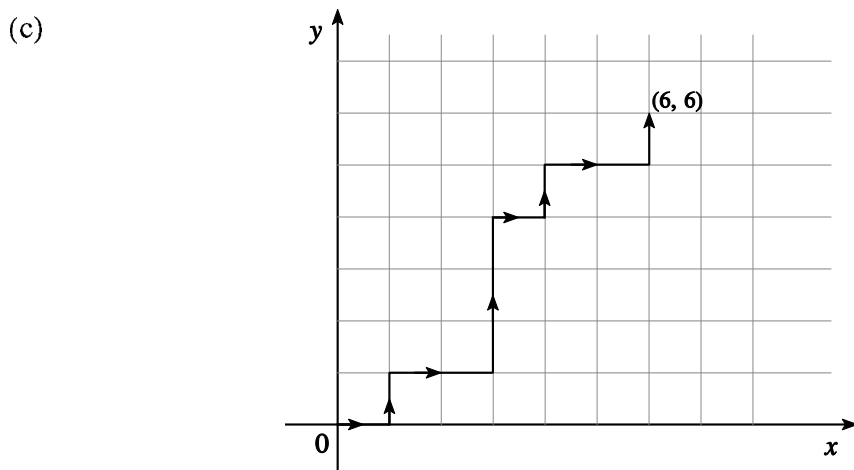
(ii) Use the result to solve the equation $8x^3 - 6x + 1 = 0$. 2

(iii) Deduce that $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$. 2

(e) $\int_0^2 \frac{dx}{\sqrt{2x-x^2}}$. [Hint: try completing the square.] 2

Question 3 (20 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $\frac{1}{1+u^2} = 1 - u^2 + u^4 - u^6 + u^8 - \dots$ 2
- (ii) By integrating both sides of the above expression from 0 to x , show 3
by a suitable choice of x , that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$
- (b) (i) If $z = x + iy$ and $z' = 1 + \frac{1}{z}$, obtain an expression for z' in the 3
form $x' + iy'$, and hence express each of x' and y' in terms of x and y .
- (ii) Find an algebraic relation between x' and y' when z has constant 3
argument θ .



The lines in the above figure represent the streets in a city. A messenger wants to get from $(0, 0)$ to (x, y) , where x, y are non-negative integers, moving in the positive directions of x and y only.

Let the number of possible routes be denoted by $f(x, y)$.

- (i) Evaluate $f(k, 1)$. 2
- (ii) Show that $f(k+1, 2) = f(k, 2) + f(k+1, 1)$. 3
- (iii) By means of mathematical induction or otherwise, prove that 4
 $f(m, 2) = \frac{1}{2}(m^2 + 3m + 2)$ for all non-negative integral values of m .

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x, \quad x > 0$

QUESTION 1.

2002 EXT 2 Solution
Task #1

$$(a) (i) |15-20i| = \sqrt{15^2 + 20^2} \\ = \sqrt{625} \\ = 25$$

$$(ii) \arg(-1 - \frac{3}{\sqrt{2}}i) = -(\pi - 1.10714) \\ \approx -2.0113 \\ \text{or } (-115.94^\circ)$$

$$(b) x^2 - 4x = -20 \\ (x-2)^2 = -16 \\ x-2 = \pm 4i \\ \underline{x = 2 \pm 4i}$$

$$(c) P(z) = z^3 - 7z^2 + 25z - 39$$

new zeroes is $2+3i$

$\therefore$ another zero is $2-3i$ (by conjugate root theorem)

new 4th root is β .

$$2+3i + 2-3i + \beta = 7 \quad (\sum z_i = -\frac{b}{a}) \\ \therefore \beta = 3$$

$$\therefore P(z) = (z-2+3i)(z-2-3i)(z-3)$$

$$\text{dx } y = x \ln x \\ y' = 2x \cdot \frac{1}{x} + 1 \cdot \ln x \\ = 1 + \ln x$$

$$\therefore \int (1 + \ln x) dx = x \ln x$$

$$\int \ln x dx = x \ln x - \int 1 \cdot dx$$

$$\therefore \int \ln x dx = x \ln x - x + c$$

$$(c) D: |x| \leq \frac{1}{2}$$

$$R: 0 \leq y \leq 3\pi$$

$$(f) \sqrt{5+12i} = a+ib \\ 5+12i = a^2 - b^2 + 2abi$$

$$\therefore \boxed{a^2 - b^2 = 5} \quad \text{--- (i)}$$

$$\boxed{2ab = 12} \quad \text{--- (ii)}$$

$$\text{new } (a^2 + b^2)^2 = (a^2 - b^2)^2 + (2ab)^2 \\ = 25 + 144 \\ = 169$$

$$\therefore \boxed{a^2 + b^2 = 13} \quad \text{--- (iii)}$$

$$\text{(i)} + \text{(iii)}$$

$$2a^2 = 18$$

$$a^2 = 9$$

$$a = \pm 3$$

$$\therefore b = \pm 2$$

Hence roots are $\pm(3+2i)$

$$\begin{aligned}
 \text{g) } P(i) &= i^3 + 2i^2 + 3i \\
 &= -i - 2i + 3i \\
 &= 0
 \end{aligned}$$

$\therefore (z-i)$ is a factor.

$$\begin{aligned}
 \text{h) Let } z_1 &= 1+6i \\
 z_2 &= 3+10i \\
 z_3 &= 4+14i
 \end{aligned}$$

$$\text{now } z_2 - z_1 = 2+4i$$

$$\therefore \arg(z_2 - z_1) = \tan^{-1} 2.$$

$$\begin{aligned}
 \text{Also } \arg(z_3 - z_2) &= \arg(1+4i) \\
 &= \tan^{-1} 4.
 \end{aligned}$$

$\therefore$ vectors $z_2 - z_1$ and $z_3 - z_2$
are parallel.

$\therefore z_2$ is on the line

$\therefore$ collinear!

$$\text{i) given } \ln x = x - 1.1.$$

$$\text{Consider } f(x) = \ln x - x + 1.1.$$

$$\text{now } f'(x) = \frac{1}{x} - 1$$

$$\begin{aligned}
 \text{If } x_1 &= 1.5 \\
 x_2 &= 1.5 - \frac{f(1.5)}{f'(1.5)}
 \end{aligned}$$

$$\doteq 1.516385$$

$$\doteq \underline{1.52} \text{ (3.S.f. figs)}$$

2

(a) If $u = \cot^{-1} x$
 then $x = \cot u$

$$\frac{du}{dx} = -\frac{1}{\sin^2 u}$$

$$\frac{dx}{du} = -\cot^2 u$$

$$\frac{d}{dx} \cot^{-1} x = \frac{-1}{\cot^2 u}$$

$$= \frac{-1}{1+x^2}$$

(d)

(1) $8x^3 - 6x = -1$

$$4x^3 - 3x = -\frac{1}{2}$$

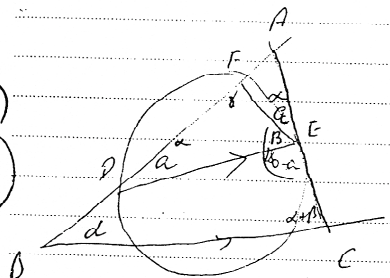
$$\cos 3\theta = -\frac{1}{2}$$

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$$

(b)

(3)



$$\angle AEF = \angle ADE = \theta$$

$$= \theta$$

$$\angle FCE = \theta - \alpha$$

opposite $\angle$'s supplementary.
 Cycle.

(2)

(1) Sum of roots $= -\frac{b}{a} = 0$

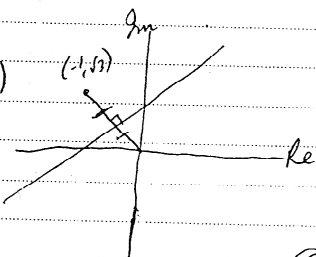
$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0$$

$$\cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}$$

$$\cos \frac{\pi}{9} + \cos \frac{5\pi}{9} = \cos \frac{\pi}{9}$$

(c) (1)

(2)



(3)

(11) $0 \leq \arg z < \frac{\pi}{4}$

$$|z| \leq 3$$

(c)

$$\int_0^2 \frac{dx}{\sqrt{2x-x^2}}$$

$$\int_0^2 \frac{dx}{\sqrt{1-(x-1)^2}}$$

$$\int_0^2 \frac{dx}{\sqrt{1-(x-1)^2}}$$

$$= \left[\sin^{-1}(x-1) \right]_0^2$$

$$= \sin^{-1} 1 - \sin^{-1} -1$$

$$= \frac{\pi}{2} - \frac{\pi}{2}$$

$$= \pi$$

(d) (1) $\cos 3\theta = (\cos \theta + i \sin \theta)^3$

$$= \cos^3 \theta + 3\cos^2 \theta i \sin \theta - 3\cos \theta \sin^2 \theta - i \sin^3 \theta$$

(3)

$$\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$$

$$= 4\cos^3 \theta - 3\cos \theta$$

$$\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$$

Question 3

(a) (i) NTP $\frac{1}{1+u^2} = 1 - u^2 + u^4 - \dots$ for $|u| < 1$ 2

RHS = $1 - u^2 + u^4 - \dots$ an infinite geometric series with $a = 1$, $r = -u^2$
 $\therefore S_\infty$ exists since $|u| < 1 \Rightarrow u^2 < 1 \therefore |r| < 1$

$$S_\infty = \frac{a}{1-r} = \frac{1}{1-(-u^2)} = \frac{1}{1+u^2} \text{ QED}$$

(ii) $\int_0^x \frac{1}{1+u^2} du = \int_0^x (1 - u^2 + u^4 - \dots) du$ 3

$$\therefore \tan^{-1} u \Big|_0^x = u - \frac{u^3}{3} + \frac{u^5}{5} - \dots \Big|_0^x$$

$$\therefore \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

Let $x = 1$

$$\therefore \tan^{-1} 1 = \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \dots \text{ QED}$$

(b) (i) $z = x + iy, z' = x' + iy' = 1 + \frac{1}{z}$ 3

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{x - iy}{x^2 + y^2}$$

$$1 + \frac{1}{z} = 1 + \frac{x - iy}{x^2 + y^2} = \left(1 + \frac{x}{x^2 + y^2}\right) - i\left(\frac{y}{x^2 + y^2}\right)$$

$$\therefore x' = 1 + \frac{x}{x^2 + y^2}, y' = -\frac{y}{x^2 + y^2}$$

(ii) $\arg z = \theta = \text{constant}$ 3

$$\tan \theta = \frac{y}{x}$$

$$x' - 1 = \frac{x}{x^2 + y^2}$$

$$-y' = \frac{y}{x^2 + y^2}$$

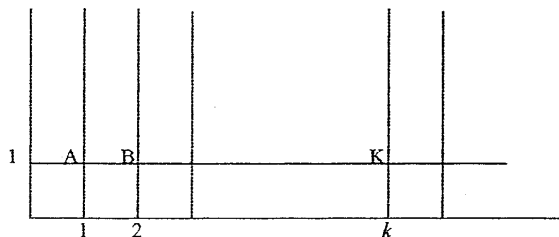
$$\therefore \frac{-y'}{x' - 1} = \frac{\frac{y}{x^2 + y^2}}{\frac{x}{x^2 + y^2}} = \frac{y}{x} = \tan \theta$$

$$\therefore \tan \theta = \frac{y'}{1 - x'}$$

(c) (i)

2

Method 1: *Addition Principle*



A corresponds to (1,1)

B corresponds to (2,1)

K corresponds to (k,1)

There is 1 way to get to (0,1) and 1 way to get to (1,0). So there is $1 + 1 = 2$ ways to get to A.
There is 1 way to get to (2,0) and 2 ways to get to (1,1). So there is $1 + 2 = 3$ ways to get to B.

So there are $k + 1$ ways to get to K.

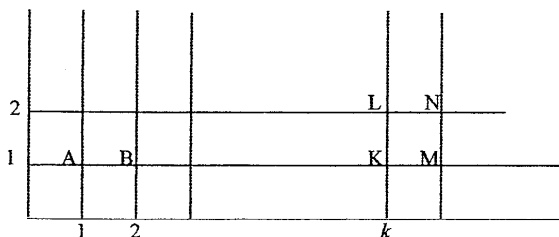
Method 2:

Use R to denote every time one move to the **RIGHT** is taken, and U to denote one move **UP**.
To get to K, there must be k R's and only 1 U. So that 1 possible route is represented by the word $\underbrace{RRR \dots RRR}_k U$. So all possible words would give all possible routes.

This means the number of permutations of $k + 1$ letters, with k letters exactly the same

$$\text{ie } \frac{(k+1)!}{k!} = k+1 = \binom{k+1}{1}$$

(ii) **Method 1:** *Addition Principle*



The number of ways to get to N is equal to the number of ways of getting to L + the number of ways of getting to M.

$f(k,2)$ = number ways to L.

$f(k+1,1)$ = number ways to M.

$f(k+2,2)$ = number ways to N.

$$\text{ie } f(k,2) + f(k+1,1) = f(k+2,2)$$

QED

3

(c) (ii) **Method 2**

$$f(k, 2) = \binom{k+2}{2}, f(k+1, 1) = \binom{k+2}{1}, f(k+1, 2) = \binom{k+3}{2}$$

$$\therefore \binom{k+2}{2} + \binom{k+2}{1} = \binom{k+3}{2}$$

because of Pascal's relationship ie $\binom{n+1}{i} = \binom{n}{i} + \binom{n}{i-1}$

(c) (iii) **Note that $m \geq 0$**

$$f(m, 2) = \binom{m+2}{2} = \frac{(m+2)!}{m!2!} = \frac{(m+2)(m+1)}{2} = \frac{1}{2}(m^2 + 3m + 2)$$

4

OR with mathematical induction

When $m = 0$, LHS = $f(0, 2) = 1$

RHS = $\frac{1}{2}(0^2 + 3 \times 0 + 2) = 1$

So true for $m = 0$

Assume true for $m = k$ ie $f(k, 2) = \frac{1}{2}(k^2 + 3k + 2)$

NTP that $f(k+1, 2) = \frac{1}{2}((k+1)^2 + 3(k+1) + 2) = \frac{1}{2}(k^2 + 5k + 6)$

LHS = $f(k+1, 2)$

$= f(k, 2) + f(k+1, 1)$ from (ii)

$= \frac{1}{2}(k^2 + 3k + 2) + (k+2)$ from (i)

$= \frac{1}{2}(k^2 + 3k + 2) + \frac{1}{2}(2k + 4)$

$= \frac{1}{2}(k^2 + 5k + 6)$

$= \text{RHS}$

So by the principle of mathematical induction, $f(m, 2) = \frac{1}{2}(m^2 + 3m + 2), m \geq 0$